60 single-best-answer questions | 90 minutes | 60 marks

Answer all questions. Choose one answer per question. Use a separate answer sheet.

1. Why does the sinoatrial (SA) node normally determine the heart rate?

A. It is the only cardiac site that can generate impulses without nerve stimulation.
B. It reaches the threshold for spontaneous discharge sooner than other pacemakers.
C. It conducts electrical impulses through ventricular muscle faster than Purkinje fibres.
D. Its cells have a more negative resting membrane potential than ventricular cells.
E. It receives stronger sympathetic input than any other pacemaker tissue.

2. What is the main benefit of slow impulse conduction through the AV node?

A. It allows the aortic valve to remain open during ventricular relaxation.
B. It allows atrial contraction to assist ventricular filling before ventricular systole.
C. It allows the pulmonary veins to contract before the atria fill.
D. It prevents the atria from depolarizing during every cardiac cycle.
E. It keeps ventricular contraction longer than ventricular relaxation at rest.

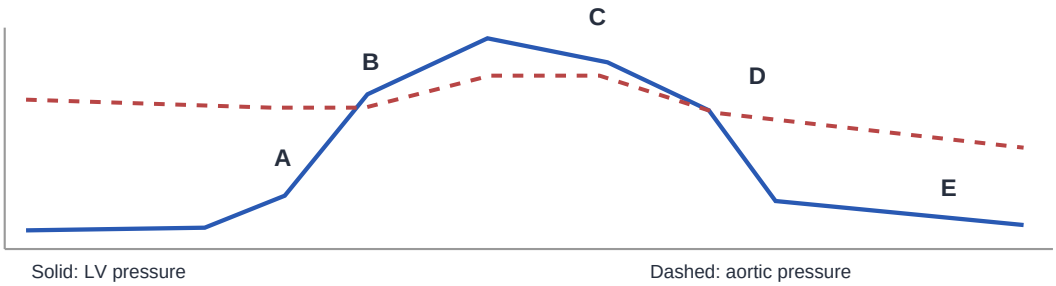
3. Left ventricular pressure has risen to 45 mmHg. Left atrial pressure is 10 mmHg and aortic pressure is 85 mmHg. Which phase is most likely?

A. Rapid ventricular filling
B. Reduced ventricular ejection
C. Isovolumetric relaxation
D. Rapid ventricular ejection
E. Isovolumetric contraction

4. Immediately after the aortic valve closes, left ventricular pressure begins to fall. Before the mitral valve opens, what happens to ventricular volume?

A. It increases quickly as blood enters from the atrium.
B. It increases slowly because the ventricle is relaxing.
C. It stays constant because both valves are closed.
D. It decreases quickly as blood leaves into the aorta.
E. It decreases slowly because the mitral valve leaks normally.

5. The diagram shows left ventricular pressure (solid) and aortic pressure (dashed). At which labelled point does aortic valve opening begin?



A. Point B
B. Point A
C. Point C
D. Point D
E. Point E

6. A ventricle has an end-diastolic volume of 140 mL and an end-systolic volume of 56 mL. What is its ejection fraction?

A. 40%
B. 60%
C. 50%
D. 65%
E. 84%

7. A pulse is felt at the wrist after the left ventricle ejects blood. What travels along the arteries to produce this pulse?

A. The same column of blood moving from the aorta to the wrist within one beat
B. A contraction wave moving through arterial smooth muscle
C. A wave of ventricular electrical activity passing along the artery
D. A pressure wave that expands the arterial wall
E. A pulse of venous pressure transmitted backward into the radial artery

8. In complete AV dissociation, an atrium sometimes contracts while the tricuspid valve is closed. Which jugular venous finding may appear?

A. A persistently absent a wave
B. Intermittent giant v waves caused by atrial systole
C. Loss of the y descent with every normal beat
D. A regular arterial-type pulse in the jugular vein
E. Intermittent giant a waves

9. Why does inspiration normally widen the split between the aortic and pulmonary components of the second heart sound?

A. Increased right ventricular filling delays mitral valve opening.
B. Increased right ventricular filling delays pulmonary valve closure.
C. Reduced right ventricular filling delays aortic valve closure.
D. Increased left ventricular filling delays tricuspid valve closure.
E. Reduced left ventricular filling delays the first heart sound.

10. A patient has backward blood flow from the left ventricle into the left atrium during systole. Which valve disorder best explains this?

A. Mitral stenosis
B. Aortic stenosis
C. Aortic regurgitation
D. Mitral regurgitation
E. Tricuspid regurgitation

11. A third heart sound (S3) is heard shortly after S2. With which part of the cardiac cycle is it associated?

A. Atrial contraction just before S1
B. Isovolumetric ventricular contraction
C. Closure of the aortic and pulmonary valves
D. Rapid ventricular filling
E. Opening of the aortic and pulmonary valves

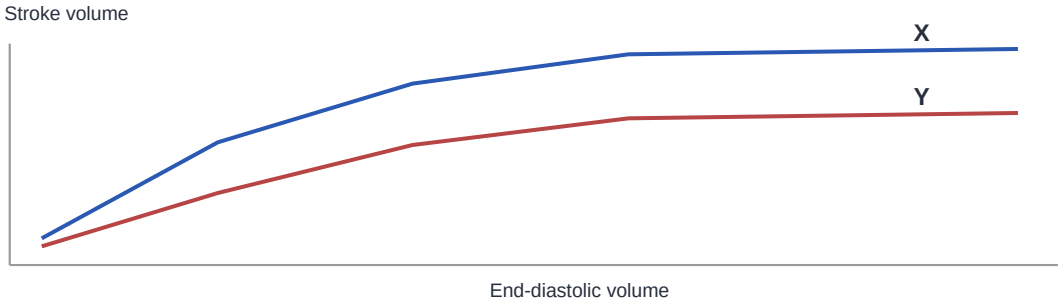
12. Venous return rises while heart rate, afterload and contractility remain unchanged. What immediate change is expected in a normal ventricle?

A. A rise in end-diastolic volume and stroke volume
B. A fall in end-diastolic volume and stroke volume
C. A rise in stroke volume with a fall in end-diastolic volume
D. A fall in stroke volume despite a rise in end-diastolic volume
E. No change in stroke volume despite greater ventricular filling

13. Aortic pressure suddenly rises. For the first affected beat, ventricular filling and contractility are unchanged. What is the likely effect on stroke volume?

A. Stroke volume increases because aortic recoil becomes stronger.
B. Stroke volume increases because ventricular filling has increased.
C. Stroke volume stays constant because heart rate has not changed.
D. Stroke volume decreases because the ventricle ejects against greater pressure.
E. Stroke volume increases because the AV node conducts more quickly.

14. Two ventricular function curves are shown. At the same end-diastolic volume, curve X produces a larger stroke volume than curve Y. What best explains the difference?



A. Ventricular preload must be greater on curve X.
B. The heart rate must be lower on curve X.
C. The aortic valve must be closed throughout ejection on curve X.
D. The ventricular muscle must be less responsive to calcium on curve X.
E. Ventricular contractility is greater on curve X.

15. At a very fast heart rate, stroke volume may fall even when the heart contracts strongly. What is the main reason?

A. Systole disappears as heart rate rises.
B. Atrial contraction stops whenever the heart rate exceeds normal.
C. The semilunar valves remain open throughout the next beat.
D. The ventricles lose all ability to contract when diastole shortens.
E. Diastole becomes too short for adequate ventricular filling.

16. Oxygen consumption is 280 mL/min. Arterial oxygen content is 200 mL/L and mixed venous oxygen content is 130 mL/L. What is cardiac output?

A. 4 L/min
B. 2 L/min
C. 3 L/min
D. 5 L/min
E. 7 L/min

17. For cardiac output measurement using the oxygen Fick principle, why is blood sampled from the pulmonary artery?

A. It contains mixed venous blood returning from the systemic circulation.
B. It contains freshly oxygenated blood leaving the lungs.
C. It measures oxygen content in the coronary arteries only.
D. It has the same oxygen content as any single peripheral vein.
E. It directly measures left ventricular stroke volume without a calculation.

18. At the same stroke volume and heart rate, which change generally increases myocardial oxygen demand more?

A. Pumping against a lower arterial pressure
B. Reducing the pressure needed to open the aortic valve
C. Pumping against a much higher arterial pressure
D. Lowering wall tension during ventricular ejection
E. Reducing the pressure difference across the aortic valve

19. According to Poiseuille's law, if a vessel radius is reduced to half while other factors remain constant, how does resistance change?

A. It becomes 2 times greater.
B. It becomes 4 times greater.
C. It becomes 16 times greater.
D. It becomes 8 times greater.
E. It becomes 16 times smaller.

20. Why is average blood-flow velocity normally much lower in capillaries than in the aorta?

A. The total capillary cross-sectional area is smaller than that of the aorta.
B. Capillaries receive a smaller total blood flow than the aorta in steady state.
C. Capillary pressure is lower, so red cells are stationary between heartbeats.
D. The combined cross-sectional area of all capillaries is much larger.
E. Blood viscosity is so much higher in capillaries that flow velocity must be zero.

21. A patient develops marked polycythaemia. If vascular diameter and pressure gradient are otherwise unchanged, what happens to resistance?

A. It decreases because blood viscosity increases.
B. It decreases because plasma oncotic pressure rises.
C. It stays unchanged because haematocrit affects oxygen only.
D. It stays unchanged because viscosity influences velocity but not flow resistance.
E. It increases because blood viscosity increases.

22. Which isolated change increases the tendency for turbulent flow in a blood vessel?

A. An increase in blood viscosity
B. A decrease in blood density
C. An increase in blood-flow velocity
D. A decrease in vessel diameter
E. A decrease in the Reynolds number

23. An electromagnetic blood-flow meter estimates flow by measuring what happens when blood moves through a magnetic field. What does it detect?

A. The oxygen consumed by blood during each passage
B. The sound produced when the blood becomes turbulent
C. An electrical voltage induced across the moving blood
D. The amount of plasma protein filtered into tissues
E. The pressure required to close a sphygmomanometer cuff

24. A resting adult has a blood pressure of 120/75 mmHg. Using the usual resting approximation, what is mean arterial pressure?

A. 75 mmHg
B. 90 mmHg
C. 85 mmHg
D. 95 mmHg
E. 105 mmHg

25. In a healthy adult, cuff pressure is slowly released during blood-pressure measurement. Disappearance of Korotkoff sounds usually marks which pressure?

A. Systolic arterial pressure
B. Mean arterial pressure
C. Diastolic arterial pressure
D. Pulse pressure
E. Central venous pressure

26. What helps maintain arterial blood flow during ventricular diastole, when the left ventricle is not ejecting?

A. Persistent ejection by the ventricle after the aortic valve closes
B. Elastic recoil of arteries stretched during systole
C. Active contraction of the aortic valve leaflets during diastole
D. Pressure transmitted from the right ventricle into the aorta
E. The skeletal muscle pump acting directly on the ascending aorta

27. A deep vein in one leg becomes obstructed, increasing pressure in vessels upstream. What local change most directly favours swelling?

A. An increase in plasma albumin concentration
B. A decrease in capillary hydrostatic pressure
C. An increase in lymphatic drainage capacity
D. A decrease in capillary wall permeability
E. An increase in capillary hydrostatic pressure

28. A patient has very low plasma albumin but unchanged capillary blood pressure. Why can oedema develop?

A. The plasma colloid osmotic force drawing fluid into vessels increases.
B. The permeability of capillaries to water decreases throughout the body.
C. The capillary hydrostatic pressure decreases enough to prevent filtration.
D. The osmotic force retaining water within the circulation is reduced.
E. Lymphatic return increases enough to oppose all local filtration.

29. A patient develops chronic swelling of an arm after axillary lymph nodes are removed. What is the main reason?

A. Arterial blood supply to the arm has completely stopped.
B. Red blood cells begin moving freely out of intact capillaries.
C. The heart pumps excess blood only into the affected arm.
D. Interstitial fluid and proteins cannot drain normally through lymph vessels.
E. Plasma albumin production increases sharply in the affected tissue.

30. Why does walking help return blood from the leg veins to the heart?

A. Muscle contractions compress veins while valves limit backward flow.
B. Venous valves in the calf actively push blood toward the heart.
C. Walking lowers all leg capillary pressures to zero.
D. Muscle compression pushes venous blood toward the feet through open valves.
E. Veins become less compliant and no longer experience gravity.

31. During exercise, working skeletal muscle produces more CO_2, H^+ and other metabolites. What local vascular response is expected?

A. Arteriolar constriction that reduces supply to active muscle
B. Arteriolar dilation that increases blood supply to the active muscle
C. Closure of all capillaries until the exercise finishes
D. A reduction in local flow despite increased oxygen demand
E. Selective closure of venous valves inside the muscle

32. Pressure inside a small artery suddenly rises. What initial response of its smooth muscle helps resist the increase in flow?

A. It relaxes in response to the extra stretch.
B. It maintains the same diameter while pressure rises.
C. It contracts in response to the extra stretch.
D. It dilates because greater pressure directly lowers smooth-muscle calcium entry.
E. It dilates because its oxygen requirement falls whenever pressure increases.

33. Blood flow through a forearm artery is briefly blocked and then restored. Why may flow temporarily rise above its resting level?

A. Vasodilator metabolites built up while the tissue was underperfused.
B. The heart has permanently increased the forearm's blood supply.
C. The arterial pulse wave has stopped travelling through the arm.
D. The blockage permanently removed vascular resistance.
E. Venous valves now actively force arterial blood toward the hand.

34. The graph shows blood flow across a range of perfusion pressures. Between points B and C, why is flow nearly constant despite rising pressure?

A. Arteriolar resistance falls as perfusion pressure rises.
B. Cardiac output stops changing throughout the entire body.
C. Arteriolar resistance rises as perfusion pressure rises.
D. Venous blood begins flowing backward into the artery.
E. The tissue stops consuming oxygen over this pressure range.

35. A person suddenly loses blood and arterial pressure falls. Which immediate autonomic change helps restore pressure?

A. Reduced sympathetic activity and increased vagal activity
B. Reduced sympathetic and reduced vagal activity together
C. Increased sympathetic activity and reduced vagal activity
D. Increased vagal activity with arteriolar dilation
E. No change in autonomic outflow until blood volume recovers

36. Arterial pressure suddenly increases. What happens to the rate of firing of carotid sinus baroreceptors?

A. It decreases because the arterial wall is stretched more.
B. It stops because baroreceptors respond only to falling pressure.
C. It increases because the arterial wall is stretched more.
D. It becomes independent of stretch and depends only on blood oxygen.
E. It is replaced completely by discharge from the carotid body chemoreceptors.

37. A patient develops rapidly rising intracranial pressure with impaired brainstem perfusion. Which combination is typical of the Cushing response?

A. Increased arterial pressure with reflex slowing of the heart
B. Decreased arterial pressure with reflex slowing of the heart
C. Increased arterial pressure with sustained reflex tachycardia
D. Decreased arterial pressure with complete loss of sympathetic tone
E. Unchanged arterial pressure with marked peripheral vasodilation

38. Rapid infusion of fluid increases venous return and stretches the right atrium. Which reflex response may help accommodate the larger venous return?

A. Heart rate may decrease because atrial stretch directly increases vagal outflow.
B. Heart rate may increase as atrial stretch activates a reflex.
C. Heart rate may remain unchanged because atrial stretch has no reflex effects.
D. Heart rate may decrease because venous return is reduced by the fluid infusion.
E. Heart rate may increase only if the arterial baroreceptors first detect hypotension.

39. Immediately after a person stands up quickly, before reflex compensation has fully developed, what tends to happen?

A. Venous pooling decreases and ventricular filling rises.
B. Venous pooling increases and venous return to the heart falls.
C. Venous pressure in the feet falls below pressure at heart level.
D. Cardiac output rises because gravity pushes blood toward the chest.
E. Cerebral blood flow increases because the head is now above the heart.

40. Why can sustained head-to-foot acceleration (+Gz) cause a pilot to lose consciousness?

A. Blood shifts toward the legs, reducing pressure and flow to the brain.
B. Blood shifts toward the head, causing cerebral venous congestion.
C. Pulmonary artery pressure becomes identical to aortic pressure.
D. The cardiac conducting system stops generating electrical impulses.
E. Cerebral arteries are forced to dilate without any change in perfusion.

41. A person loses a substantial amount of blood. What sequence most directly explains the initial fall in cardiac output?

A. Lower blood volume directly raises ventricular end-diastolic volume.
B. Lower blood volume immediately increases plasma oncotic pressure enough to raise output.
C. Lower blood volume causes the aortic valve to open earlier at a higher stroke volume.
D. Lower blood volume leads to lower venous return and ventricular filling.
E. Lower blood volume raises pulmonary venous return above systemic venous return.

42. A patient with severe infection has widespread arteriolar dilation and low arterial pressure. Which shock mechanism is most likely?

A. Hypovolaemic shock caused mainly by external blood loss
B. Cardiogenic shock caused mainly by poor ventricular pumping
C. Obstructive shock caused mainly by mechanical blockage of blood flow
D. A normal circulation with a temporary rise in vascular resistance
E. Distributive shock caused mainly by reduced systemic vascular resistance

43. During prolonged severe shock, lactic acidosis worsens and the circulation becomes less responsive to catecholamines. What may happen next?

A. Tissue oxygen delivery automatically returns to normal.
B. Systemic vascular resistance must remain high indefinitely.
C. Myocardial function improves because acid increases calcium sensitivity.
D. Vascular tone and cardiac performance can deteriorate further.
E. Renal blood flow becomes independent of arterial pressure.

44. A patient with left ventricular failure develops breathlessness and pulmonary oedema. What pressure change most directly promotes fluid movement into the lungs?

A. Increased pulmonary venous and capillary hydrostatic pressure
B. Decreased pulmonary venous pressure
C. Increased plasma oncotic pressure in pulmonary capillaries
D. Decreased hydrostatic pressure in pulmonary capillaries
E. Increased lymphatic drainage with unchanged capillary pressure

45. A student faints after seeing a frightening injury. Immediately before fainting, which combination is most consistent with a vasovagal episode?

A. Increased heart rate and widespread vasoconstriction
B. Increased heart rate with marked skeletal-muscle vasoconstriction only
C. Unchanged heart rate with increased cardiac contractility
D. Increased blood volume and increased venous return
E. Reduced heart rate and widespread vasodilation

46. An obstruction mainly affects blood supply to the lateral surface of a cerebral hemisphere. Which major cerebral artery is most likely involved?

A. Anterior cerebral artery
B. Posterior cerebral artery
C. Anterior communicating artery
D. Middle cerebral artery
E. Basilar artery

47. The cerebral aqueduct becomes blocked while CSF production continues. Which ventricles are expected to enlarge first?

A. The fourth ventricle alone
B. The third and fourth ventricles only
C. The lateral ventricles and fourth ventricle only
D. The lateral and third ventricles
E. The fourth ventricle and central canal only

48. Brain capillary endothelial cells have tight junctions. How does glucose normally cross into brain tissue?

A. By moving freely between endothelial cells despite their tight junctions
B. By passing directly through membrane lipid without any carrier
C. By crossing endothelial ion channels that transport sodium and potassium
D. By passing from plasma to astrocytes without crossing the capillary endothelium
E. Through specific glucose transport proteins in the endothelial membrane

49. Mean arterial pressure is 95 mmHg and intracranial pressure is 15 mmHg. Assuming cerebral venous pressure is not higher than intracranial pressure, what is cerebral perfusion pressure?

A. 65 mmHg
B. 75 mmHg
C. 80 mmHg
D. 95 mmHg
E. 110 mmHg

50. Why does most left ventricular coronary blood flow occur during diastole rather than systole?

A. The left coronary artery is closed by the aortic valve during systole.
B. Blood in coronary arteries contains oxygen only during diastole.
C. The heart stops using oxygen during systole.
D. Pulmonary venous pressure becomes higher than aortic pressure during systole.
E. Contracting left ventricular muscle compresses vessels within its wall.

51. A region of heart muscle works harder and uses more oxygen. Which local change helps increase coronary blood flow?

A. Build-up of metabolites such as adenosine promotes arteriolar dilation.
B. Reduced metabolic activity causes arteriolar dilation in that region.
C. Increased oxygen extraction alone is always sufficient at any workload.
D. Aortic pressure becomes independent of coronary vessel resistance.
E. Increased adenosine produces sustained constriction of the local coronary arterioles.

52. One group of alveoli is poorly ventilated while nearby alveoli are well ventilated. What happens to the small pulmonary arteries supplying the poorly ventilated region?

A. They dilate, directing more blood toward the poorly ventilated alveoli.
B. They remain unchanged because oxygen never affects pulmonary resistance.
C. They constrict only if systemic arterial pressure is also high.
D. They close because all pulmonary arteries normally lack smooth muscle.
E. They constrict, directing more blood toward better-ventilated alveoli.

53. During exercise, pulmonary blood flow increases greatly but pulmonary arterial pressure rises relatively little. What helps explain this?

A. Most pulmonary capillaries close as flow increases.
B. Previously underperfused capillaries open and existing vessels widen.
C. Pulmonary vascular resistance increases in direct proportion to flow.
D. The right ventricle stops pumping blood during exercise.
E. The alveoli stop exchanging gases while blood flow rises.

54. During severe haemorrhage, sympathetic vasoconstriction reduces blood flow to the gut. What is one immediate benefit of this response?

A. It increases digestion by delivering more blood to intestinal villi.
B. It helps preserve perfusion pressure for vital organs by reducing splanchnic flow.
C. It prevents any reduction in cardiac output regardless of blood loss.
D. It sends all oxygenated blood directly into the hepatic portal vein.
E. It permanently prevents ischemic damage to the intestinal mucosa.

55. After firm stroking of the skin, redness spreads beyond the directly stimulated line. What produces this surrounding flare?

A. An axon reflex causing nearby arterioles to dilate
B. Clotting of capillaries at the site of injury
C. A sudden fall in total circulating blood volume
D. Closure of all precapillary sphincters around the area
E. Direct filtration of red blood cells into the epidermis

56. Fetal haemoglobin binds oxygen more strongly than adult haemoglobin under comparable conditions. Which property contributes to this difference?

A. Local fluid filtration that forms the raised wheal
B. Dilation restricted to vessels directly under the stroked line
C. Reflex withdrawal of sympathetic tone from the entire skin circulation
D. Venous congestion caused by pressure on deep veins beneath the skin
E. Fetal haemoglobin binds 2,3-BPG less strongly.

57. What change occurs in the pulmonary circulation with the newborn's first breaths?

A. Pulmonary vascular resistance rises as alveolar oxygen rises.
B. Pulmonary blood flow stops because placental blood flow has ended.
C. Pulmonary vascular resistance falls as the lungs expand and oxygen rises.
D. The ductus venosus begins carrying blood from the right ventricle.
E. Systemic vascular resistance falls because the placenta remains connected.

58. Which pair of electrodes forms standard bipolar lead I in an ECG?

A. Right arm (negative) and left leg (positive)
B. Left arm (negative) and left leg (positive)
C. Left leg (negative) and right arm (positive)
D. Right arm (negative) and left arm (positive)
E. Left arm (negative) and right arm (positive)

59. An ECG has a PR interval of 0.24 seconds, with a normal-width QRS complex and every P wave followed by a QRS. What does this pattern indicate?

A. First-degree AV block (prolonged AV conduction)
B. Complete AV block with no atrioventricular conduction
C. A shortened PR interval with normal conduction
D. Ventricular fibrillation with absent organised QRS complexes
E. Atrial fibrillation with irregular atrial activation

60. In the ECG strip, the distance between consecutive R waves is 20 small squares. The paper speed is 25 mm/s. What is the heart rate?

A. 60 beats/min
B. 75 beats/min
C. 80 beats/min
D. 100 beats/min
E. 120 beats/min